QDArchive

Part 2 - Classification Report

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This report summarizes project type assignment, ISIC Rev. 5 classification, repository-wise results, and limitations for the QDArchive Part 2 workflow.

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Executive Summary

This report presents the Part 2 classification results for the QDArchive seeding project. The database contains 5936 projects and 114763 file records from two repositories: DANS and ICPSR. The goal of this phase was to assign every project a project type and to classify QDA_PROJECT and QD_PROJECT entries using the ISIC Rev. 5 division-level taxonomy.

The project type filtering identified 7 QDA_PROJECT entries and 4570 QD_PROJECT entries in repository_id=5 (DANS). These entries were classified at project level and primary-data-file level. In total, 68509 classification targets were prepared and 68509 classification results were stored in the SQLite database.

Repository_id=15 (ICPSR) is present in the database, but its 132 projects were classified as NOT_A_PROJECT. This happened because the ICPSR acquisition stored metadata-only file records such as restricted or login-required entries, not concrete primary data file extensions. Therefore, ICPSR rows are included in the final XLSX table, but they do not have ISIC primary or secondary classes.

Metric	Value
Total projects	5936
Total file records	114763
Classification targets	68509
Classification results	68509
Repositories	DANS, ICPSR
ISIC level used	Division level

Methodology

Project type assignment

Each project was first assigned to one of four project types using the file extensions stored in the files table. A project was labelled QDA_PROJECT when at least one qualitative data analysis file extension was found. If no QDA file was found but at least one primary data extension was present, the project was labelled QD_PROJECT. Projects with other valid data file types were labelled OTHER_PROJECT. Projects without usable file-type evidence were labelled NOT_A_PROJECT.

Classification target preparation

For QDA_PROJECT and QD_PROJECT entries, two levels of classification input were created. First, a project-level target was created using the project title, description, language, keywords, file types, and file names. Second, a file-level target was created for each primary data file. This followed the requirement to classify both the project as the sum of its files and each primary data file.

ISIC Rev. 5 taxonomy import

The ISIC Rev. 5 taxonomy was imported from the provided Excel file. The classifier used the hierarchy down to division level, resulting in 87 division classes. These division descriptions became the reference documents for automatic classification.

TF-IDF similarity classifier

The classifier used TF-IDF vectorization with English stop-word removal and unigram/bigram features. Each project or file target was compared against the ISIC division descriptions using cosine similarity. The highest-scoring division was stored as the primary class. A secondary class was stored only when the second-best match was close enough to the primary score.

Repository-wise analysis

The results were analyzed by repository and by project type. Histograms and top-20 tables were generated for classified project-level results. OTHER_PROJECT and NOT_A_PROJECT entries were not classified with ISIC, but they remain visible in the database and in the XLSX export for completeness.

Overall Project Type Distribution

Repository	Repository ID	Project type	Count
DANS	5	NOT_A_PROJECT	7
DANS	5	OTHER_PROJECT	1220
DANS	5	QDA_PROJECT	7
DANS	5	QD_PROJECT	4570
ICPSR	15	NOT_A_PROJECT	132

Interpretation

The overall distribution shows that repository_id=5 contains the useful classification material for this phase. Most DANS projects were labelled QD_PROJECT because they contained primary data file extensions. A smaller number of DANS projects were labelled QDA_PROJECT because they contained QDA file extensions. Repository_id=15 contains only NOT_A_PROJECT rows because the collected ICPSR file records were metadata-only and did not provide concrete downloaded primary data file extensions.

Classification Target Summary

Repository	Project type	Target level	Count
DANS	QDA_PROJECT	file	170
DANS	QDA_PROJECT	project	7
DANS	QD_PROJECT	file	63762
DANS	QD_PROJECT	project	4570

Interpretation

The target table confirms that the classifier was run on both project-level and file-level targets for QDA_PROJECT and QD_PROJECT entries. This follows the project instruction to classify the project itself and each primary data file. Since ICPSR did not contain QDA_PROJECT or QD_PROJECT entries after file-type filtering, it does not appear in the classification target table.

Repository Analysis: DANS (repository_id=5)

Repository project type overview

Project type	Count
NOT_A_PROJECT	7
OTHER_PROJECT	1220
QDA_PROJECT	7
QD_PROJECT	4570

Repository explanation

DANS is the main repository that produced usable project-level and file-level classification results. The DANS data contains both QDA_PROJECT and QD_PROJECT entries. Therefore, ISIC classification was applied to the DANS projects and their primary data files. The large number of QD_PROJECT entries indicates that many projects contained primary qualitative data files such as text documents, PDFs, spreadsheets, images, or similar file types.

Classification analysis for QDA_PROJECT

The dominant primary class for DANS QDA_PROJECT entries is O80 - Investigation and security activities with 2 project-level classifications.

Top 20 primary ISIC classes: DANS - QDA_PROJECT

Rank-ordered top 20 classes

Rank	Primary class	Class name	Count
1	O80	Investigation and security activities	2
2	E36	Water collection, treatment and supply	1
3	M68	Real estate activities	1
4	N75	Veterinary activities	1
5	O77	Rental and leasing activities	1
6	T94	Activities of membership organizations	1

Analysis and comments

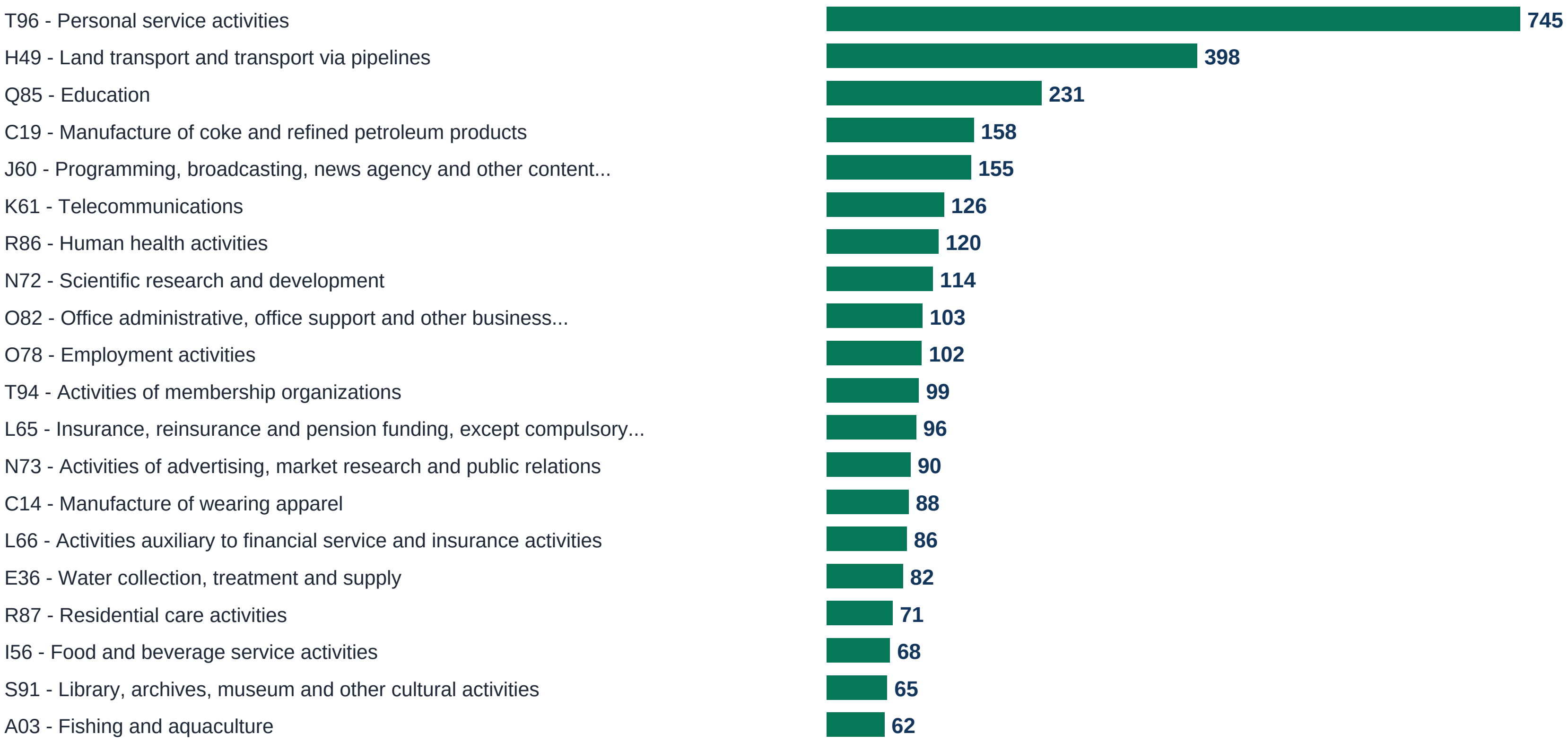
The QDA_PROJECT group is very small, with only seven projects. Because of this small sample size, the dominant class should be interpreted carefully. A small number of projects can strongly influence the class distribution. The result is still useful because it verifies that QDA projects can be detected and classified separately from ordinary qualitative data projects.

The TF-IDF method is transparent and reproducible, but it has limitations. It does not understand context as deeply as a supervised or embedding-based model. It assigns classes based on textual similarity between project metadata and ISIC division descriptions. Therefore, projects with short descriptions, missing keywords, or generic file names may receive weak or unexpected ISIC matches.

Classification analysis for QD_PROJECT

The dominant primary class for DANS QD_PROJECT entries is T96 - Personal service activities with 745 project-level classifications.

Top 20 primary ISIC classes: DANS - QD_PROJECT



Rank-ordered top 20 classes

Rank	Primary class	Class name	Count
1	T96	Personal service activities	745
2	H49	Land transport and transport via pipelines	398

Rank	Primary class	Class name	Count
3	Q85	Education	231
4	C19	Manufacture of coke and refined petroleum products	158
5	J60	Programming, broadcasting, news agency and other content distribution activities	155
6	K61	Telecommunications	126
7	R86	Human health activities	120
8	N72	Scientific research and development	114
9	O82	Office administrative, office support and other business support activities	103
10	O78	Employment activities	102
11	T94	Activities of membership organizations	99
12	L65	Insurance, reinsurance and pension funding, except compulsory social security	96
13	N73	Activities of advertising, market research and public relations	90
14	C14	Manufacture of wearing apparel	88
15	L66	Activities auxiliary to financial service and insurance activities	86
16	E36	Water collection, treatment and supply	82
17	R87	Residential care activities	71
18	I56	Food and beverage service activities	68
19	S91	Library, archives, museum and other cultural activities	65
20	A03	Fishing and aquaculture	62

Analysis and comments

The QD_PROJECT group is much larger and therefore gives a broader view of the repository content. The dominant classes reflect the textual signals available in project titles, descriptions, keywords, and file names. Since many records contain generic metadata or file names, the classifier results should be interpreted as automatic first-pass labels rather than manually verified labels.

The TF-IDF method is transparent and reproducible, but it has limitations. It does not understand context as deeply as a supervised or embedding-based model. It assigns classes based on textual similarity between project metadata and ISIC division descriptions. Therefore, projects with short descriptions, missing keywords, or generic file names may receive weak or unexpected ISIC matches.

Repository Analysis: ICPSR (repository_id=15)

Repository project type overview

Project type	Count
NOT_A_PROJECT	132

Repository explanation

ICPSR is present in the database and in the final XLSX export, but it was not classified with ISIC. All ICPSR projects were labelled NOT_A_PROJECT because the collected file records were metadata-only. The file status values show restricted or login-required records, meaning there was not enough file-type evidence to derive QDA_PROJECT or QD_PROJECT labels.

ICPSR file status evidence

File type	Status	Count
metadata_only	RESTRICTED	113
metadata_only	FAILED_LOGIN_REQUIRED	19

Classification analysis

No ISIC primary-class histogram or top-20 class table is shown for this repository because no project from this repository was eligible for ISIC classification after project type filtering. This should not be interpreted as missing repository data; it means the projects were outside the classifier scope.

Analysis and comments

The ICPSR result is important for the final interpretation because it demonstrates the effect of restricted or metadata-only acquisition. The repository was successfully represented in the database, but the lack of downloaded files prevented file-extension-based identification of qualitative data projects. For future work, improving authentication handling or obtaining accessible file metadata from ICPSR would likely change some of these NOT_A_PROJECT labels into QD_PROJECT or QDA_PROJECT labels.

Limitations

Limitation	Explanation
Low metadata detail	Some projects contain short or generic titles, descriptions, keywords, or file names. This reduces classifier confidence.
Metadata-only ICPSR records	ICPSR records were present, but they were restricted or login-required and therefore did not provide usable file-type evidence.
Automatic classification	The ISIC labels were produced automatically and were not manually validated.
TF-IDF matching	TF-IDF gives reproducible similarity scores, but it does not capture deep semantic meaning.
Secondary classes	Secondary classes were stored only when the second-best class was close enough to the primary class.

Conclusion

The Part 2 workflow successfully added project type labels, imported the ISIC Rev. 5 taxonomy, created classification targets, ran a division-level classifier, exported the required XLSX table, and created the final classification database. The DANS repository produced both QDA_PROJECT and QD_PROJECT entries and was classified with ISIC. The ICPSR repository remained visible in the outputs but was not classified because all its projects were labelled NOT_A_PROJECT based on available file-type evidence.

Overall, the result provides a reproducible first-pass classification pipeline for QDArchive seeding. The database and exported files can be used for further analysis, manual validation, and future improvement of the classifier.